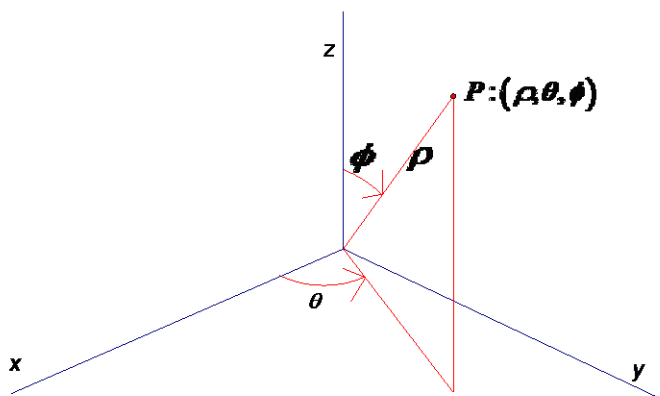


4.6.2 Triple Integrals in Spherical Coordinates

Like cylindrical coordinates, the spherical coordinate system is used to make certain triple integrals easy to evaluate. A picture of a point given by its spherical coordinates is:



When trying to derive equations that relate rectangular and spherical coordinates we obtain the following right triangle from the above picture:

To Convert from Spherical to Rectangular Coordinates:

Use the following equations: $x = \rho \sin(\phi) \cos(\theta)$, $y = \rho \sin(\phi) \sin(\theta)$, and $z = \rho \cos(\phi)$.

Remark: We restrict ρ so that $\rho \geq 0$, and we restrict ϕ so that $0 \leq \phi \leq \pi$. This means that

$$\rho = \sqrt{x^2 + y^2 + z^2} \Leftrightarrow \rho^2 = x^2 + y^2 + z^2.$$

Example 1: Describe the graph of the following equation in spherical: $\phi = \frac{\pi}{6}$.

Fact: Suppose that $E = \{(\rho, \theta, \phi) : \alpha \leq \theta \leq \beta, c \leq \phi \leq d, g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi)\}$ where $\beta - \alpha \leq 2\pi$, $d - c \leq \pi$, and both $g_1(\theta, \phi)$ and $g_2(\theta, \phi)$ are continuous and nonnegative for all possible values of θ and ϕ . Then, assuming that $f(x, y, z)$ is continuous, we have that

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_c^d \int_{g_1(\theta, \phi)}^{g_2(\theta, \phi)} f(x, y, z) \rho^2 \sin(\phi) d\rho d\phi d\theta.$$

Note: The above formula is partially justified on pages 1051 and 1052 of the textbook.

Note: Since theta and phi are bounded by constants their order can be reversed.

Note: The general procedure for setting up triple integrals as iterated integrals does not extend in a natural way to spherical since, currently; it doesn't make sense to project onto the $\phi\theta$ -plane (this will be revisited in section 4.7). We will therefore take a more intuitive approach.

Example 2: Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$ where $B = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$.

Example 3: Find the exact volume of the solid that lies inside the sphere $x^2 + y^2 + z^2 = 4$, above the xy -plane, and below the half-cone $z = \sqrt{x^2 + y^2}$.

